Lab Procedure

Weather Sensor

Introduction

Ensure the following:

1. You have reviewed the Application Guide – Weather Sensor
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
    a. The LED next to the USB C port will stay white after the touch screen starts loading.
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Measuring Weather Signals

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**…, and browse to the directory containing the python files and lab procedure for the Weather Sensor lab (…/sensors_trainer/1_fundamentals/weather_sensor/hardware/python). Open this directory.
2. Open weather.py. The device is initialized in the line:

    ```
    with SensorsTrainer() as sensors:
    ```

3. The frequency at which data will be read (10Hz) and the amount of time which the code will be running at are defined at the top of the file using the **`frequency`** and **`simulationTime`** variables. The code will run at a frequency of 10Hz as defined by the variable **`frequency`**, and for a total of 300 seconds as defined by the variable **`simulationTime`**.
4. Run weather.py using the ▷ button or the terminal. This will open 4 plots **`Temperature, Pressure, Humidity`** and **`Altitude`**. Close the Altitude Scope for now, it will be used later in the lab. Move the scopes so you can see them all.
5. What are the units of measurement from this sensor for each of the 3 measurements **`Temperature, Pressure`** and **`Humidity`**?
6. Lightly blow a puff of air into the weather sensor from somewhere around 5 cm (2 in) away from the device. How do these signals change? Do all of them return to their previous measurements at the same time? If not, which one is the slowest and the fastest? Try it a couple of times.

7. Grab your device away from the weather sensor (from the bottom or the light resistor side) so your body temperature doesn't affect readings. Bring it lower to the ground and then higher above your head. Do any of these signals vary much? If so, how? Do they consistently go higher or lower depending on the scenario? Does the speed at which you move the device affect it?

8. Click back on the terminal in Visual Studio Code and stop the script using **`Ctrl+C`** and clicking enter when prompted (the script stops after 300 seconds automatically).

Measuring Altitude

The pressure altitude is the altitude predicted by the International Standard Atmosphere (ISA) based on a measured atmospheric pressure. The equation to convert atmospheric pressure in millibars (mb) to pressure altitude (h) in feet (ft), is the following:

$$h_{\mathrm{ft}} = 145366.45\left[1 - \left(\frac{P_{\mathrm{mb}}}{P_0}\right)^{0.190284}\right]$$

Where h_{ft} is the pressure altitude in feet, P_0 is the pressure at sea level, which is 1013.25 mb, and P is the measured atmospheric pressure in millibars.

9. Considering that the sensor reading gives pressure in Pascals (Pa), how could you get altitude in both feet and meters based on the above formula?

10. Search online for the elevation above sea level in your city or as close to you as you can.

11. Search online for the atmospheric pressure in your city or as close to you as you can. Sometimes the weather websites will report that value.

12. Replace the **`altitude`** variable in the weather.py file with the correct formula to convert from pascals into meters.

13. Run weather.py using the ▷ button or the terminal.

14. Compare the **`altitude`** and **`pressure`** data you are getting with the values you found online for your location? Are they the same or different? If they are different, by how much?

15. Look at your **`altitude`** plot, lift the device up (around 1 meter) and look at the reading, does it read the difference in distance properly? Would you use this device and reading for an absolute or relative altitude change? Bring the device back to your desk.

16. Repeat the previous step, this time bringing the device lower towards the floor (around 1 meter).

17. Click back on the terminal in Visual Studio Code and stop the script using **`Ctrl+C`** and clicking enter when prompted (the script stops after 300 seconds automatically).

18. Open the datasheet of the sensor BME280 located in the weather sensor folder (…/sensors_trainer/1_fundamentals /weather_sensor) and read the second page that has the key information about the sensor.

19. Based on the operating range for the pressure, what would be the altitudes at which this sensor works?

20. Report the response time under the datasheet parameters for the humidity sensor. How does this value relate to your observed response during the first part of the experiment?

21. Based on section 1 of the datasheet starting in page 8 (1. Specification), what are the resolutions of each of the sensors? What does the resolution tell us about the sensor?

22. Stop, save and close your program.

23. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.